

Topic16: Fetch API (To call to serverside api's)

1. Eclipse download and setup for the development

2. Setup the serverside Spring Boot Application

<https://start.spring.io>

dependencies:

- Spring Web WebBuild web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

application.properties:

spring.application.name=app1

server.port=9090

3. Create React project

npx create-react-app app16

4. Give a call to serverside API.

API:

@RestController

@RequestMapping("test")

@CrossOrigin

public class TestController

{ @GetMapping("t1")

public void test1() {

System.out.println("I am from test1");

}

}

React Component:

export default function Test1()

{ const callToApi = () =>{

fetch('http://localhost:9090/test/t1')

.then(() => console.log('ok'))

.catch(() => console.log('error'))

}

return (<>

<button onClick={callToApi} type='button'>Call to API</button>

</>)

}

5. Get the raw message from the server

API:

@RestController

@RequestMapping("test")

@CrossOrigin

public class TestController

{ @GetMapping("t2")

public String test2() {

System.out.println("I am from test2");

return "Hello to all from API";

}

}

React Component:

export default function Test2()

{ const callToAPI = () =>{

fetch('http://localhost:9090/test/t2')

.then(response => console.log(response))

.catch(error => console.log(error))

```

    }
    return (<
      <button onClick={callToAPI}>Call to API</button>
    </>)
  }
}

```

6. Get a message from a server but inside a JSON object

```

public class Test {
    private String message;
    public void setMessage(String message)
        { this.message = message;
    }
    public String getMessage() {
        return message;
    }
}

@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController
{ @GetMapping("t3")
  public Test test3() {
      System.out.println("I am from test3");
      Test obj = new Test();
      obj.setMessage("I am from the server side API");
      return obj;
  }
}

```

React Component:

```

export default function
Test3(){ const callToApi = ()
=>{
  fetch('http://localhost:9090/test/t3')
  .then(response => response.json())
  .then(data =>
    { console.log(data.message);
  })
  .catch(error => console.log(error))
}
return (<
  <button onClick={callToApi} type='button'>Call To API</button>
</>)
}

```

7. Get a raw array from the server

API:

```

@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController
{ @GetMapping("t4")
  public String[] test4() {
      System.out.println("I am from test4");
      String[] names = {"ramu", "test", "hello"};
      return names;
  }
}

```

React Component:

```

export default function Test4()
{ const callToAPI = () =>{
  fetch('http://localhost:9090/test/t4')
  .then(response => console.log(response))
  .catch(error => console.log(error))
}
return (<
  <button onClick={callToAPI}>Call to API</button>
</>)
}

```

8. Get an array content but inside a JSON object

API:

```
public class Test1 {
    private String[] names;
    public void setNames(String[] names)
        { this.names = names;
    }
    public String[] getNames() {
        return names;
    }
}

@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController
{ @GetMapping("t5")
  public Test1 test5() {
      System.out.println("I am from test5");
      Test1 obj = new Test1();
      String[] names = {"ramu", "test", "hello"};
      obj.setNames(names);
      return obj;
  }
}
```

React Component:

```
import { useState } from "react";
export default function Test5(){
    const [msg, setMsg] = useState('');
    const callToApi = () =>{
        fetch('http://localhost:9090/test/t5')
        .then(response => response.json())
        .then(data =>
            { console.log(data.names)
              ; setMsg(data.names);
            })
        .catch(error => console.log(error))
    }
    return (<
        <h1>{msg.join(', ')}</h1>
        <button onClick={callToApi} type='button'>Test5</button>
    </>)
}
```

9. Get list content inside a JSON object

API:

```
package com.example.app1;
import java.util.List;
public class Test2 {
    private List<Integer> list;
    public void setList(List<Integer> list)
        { this.list = list;
    }

    public List<Integer> getList() {
        return list;
    }
}

@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController
{ @GetMapping("t6")
  public Test2 test6() {
      System.out.println("I am from test6");
      Test2 obj = new Test2();
      List<Integer> list = new ArrayList<Integer>();
      list.add(100);
      list.add(200);
      list.add(400);
  }
}
```

```

        list.add(600);
        obj.setList(list);
        return obj;
    }
}

```

React Component:

```

import { useState } from "react";
export default function Test6(){
    const [values, setValues] = useState([]);
    const callToApi = () =>{
        fetch('http://localhost:9090/test/t6')
        .then(response => response.json())
        .then(data =>
            { console.log(data.list);
              setValues(data.list);
            })
        .catch(error => console.log(error))
    }
    return (<
        <h1>{values.join(', ')}</h1>
        <button onClick={callToApi} type='button'>Test6</button>
    </>)
}

```

10. Get set content inside a JSON object

Do by yourself as it's a similar to List

11. Get map content inside a JSON object

API:

```

import java.util.Map;
public class Test3 {
    private Map<String, String> educations;

    public void setEducations(Map<String, String> educations)
    { this.educations = educations;
    }

    public Map<String, String> getEducations()
    { return educations;
    }
}

@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController
{ @GetMapping("t7")
  public Test3 test7() {
      System.out.println("i ma in test3");
      Test3 obj = new Test3();
      Map<String, String> map = new HashMap<String, String>();
      map.put("ug", "B.Tech");
      map.put("pg", "M.Tech");
      obj.setEducations(map);
      return obj;
  }
}

```

React component:

```

import { useState } from "react";
export default function Test7(){
    const [values, setValues] = useState({});
    const callToApi = () =>{
        fetch('http://localhost:9090/test/t7')
        .then(response => response.json())
        .then(data =>
            { console.log(data.educations);

```

```

        setValues(data.educations);
    })
    .catch(error => console.log(error))
}
return (<
  <h1>{Object.entries(values).map([(key, value)] =>
    <p key={key}>{key}: {value}</p>)}</h1>
    <button onClick={callToApi} type='button'>Test7</button>
  </>)
)

```

12. Get Person object inside a JSON object

API:

```

package com.example.app1;
public class Person {
    private String firstName;
    private String lastName;
    private Integer age;
    public String getFirstName() {
        return firstName;
    }
    public void setFirstName(String firstName)
        { this.firstName = firstName;
    }
    public String getLastName() {
        return lastName;
    }
    public void setLastName(String lastName)
        { this.lastName = lastName;
    }
    public Integer getAge() {
        return age;
    }
    public void setAge(Integer age) {
        this.age = age;
    }
}

@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController
{
    @GetMapping("t8")
    public Person test8() {
        Person p1 = new Person();
        p1.setFirstName("abc");
        p1.setLastName("xyz");
        p1.setAge(22);
        return p1;
    }
}

```

React Component:

```

import { useState } from "react";
export default function Test8(){
    const [person, setPerson] = useState({});
    const callToApi = () =>{
        fetch('http://localhost:9090/test/t8')
        .then(response => response.json())
        .then(data =>
            { console.log(data);
              setPerson(data);
            })
        .catch(error => console.log(error))
    }
    return (<
        <h1>First Name : {person.firstName}</h1>
        <h1>Last Name : {person.lastName}</h1>
        <h1>Age : {person.age}</h1>
        <button onClick={callToApi} type='button'>Call to API</button>
    </>)
}

```

```
}
```

13. Get array of Person objects inside a JSON object API:

```
public class Family {  
    private Person[] members;  
    public Person[] getMembers() {  
        return members;  
    }  
    public void setMembers(Person[] members) {  
        this.members = members;  
    }  
}
```

```
@RestController  
@RequestMapping("test")  
@CrossOrigin  
public class TestController  
{ @GetMapping("t9")  
    public Family test9() {  
        Person p1 = new Person();  
        p1.setFirstName("abc");  
        p1.setLastName("xyz");  
        p1.setAge(22);  
  
        Person p2 = new Person();  
        p2.setFirstName("test1");  
        p2.setLastName("hello");  
        p2.setAge(24);  
  
        Person p3 = new Person();  
        p3.setFirstName("ramu");  
        p3.setLastName("rao");  
        p3.setAge(23);  
  
        Person[] persons = {p1, p2, p3};  
        Family obj = new Family();  
        obj.setMembers(persons);  
        return obj;  
    }  
}
```

React Component:

```
import { useState } from "react";  
import Test9Child from "./Test9Child";  
export default function Test9(){  
    const [persons, setPersons] = useState([]);  
    const callToApi = () =>{  
        fetch('http://localhost:9090/test/t9')  
        .then(response => response.json())  
        .then(data => { console.log(data.members);  
            setPersons(data.members);  
        })  
        .catch(error => console.log(error))  
    }  
    return (<>  
        {persons.map(person =><Test9Child person={person}/>)}  
        <button onClick={callToApi} type='button'>Call to API</button>  
    </>)  
}
```

```
export default function Test9Child({person})  
{ return (<>  
    <h1>First Name : {person.firstName}</h1>  
    <h1>Last Name : {person.lastName}</h1>
```

```

    <h1>Age : {person.age}</h1>
    <hr />
  </>
}

```

14. Get multiple Person objects inside a HashMap. This HashMap is in the JSON object

API:

```

package com.example.app1;
import java.util.Map;
public class SimpleFamily {
    private Map<String, Person> members;
    public void setMembers(Map<String, Person> members)
        { this.members = members;
    }
    public Map<String, Person> getMembers()
        { return members;
    }
}

@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController
{
    @GetMapping("t10")
    public SimpleFamily test10() {
        Person p1 = new Person();
        p1.setFirstName("abc");
        p1.setLastName("xyz");
        p1.setAge(22);

        Person p2 = new Person();
        p2.setFirstName("test1");
        p2.setLastName("hello");
        p2.setAge(24);

        Person p3 = new Person();
        p3.setFirstName("ramu");
        p3.setLastName("rao");
        p3.setAge(23);

        Map<String, Person> map = new HashMap<String, Person>();
        map.put("husband", p1);
        map.put("wife", p2);
        map.put("child", p3);

        SimpleFamily obj = new SimpleFamily();
        obj.setMembers(map);

        return obj;
    }
}

```

React Component:

```

import { useState } from "react"
import Test9Child from "./Test9Child"
export default function Test10() {
    const [persons, setPersons] = useState({})
    const callToApi = () => {
        fetch('http://localhost:9090/test/t10')
        .then(response => response.json())
        .then(data => setPersons(data.members))
        .catch(error => console.log(error))
    }
    return (<
        {Object.entries(persons).map([[key, value]] =><
            {key}: <Test9Child person = {value}/></>
        }
        <button onClick={callToApi}>Get Data</button>
    </>)
}

```

15. Get a List which contains multiple HashMaps.. Return List in the JSON object

API:

```
package com.example.app1;
import java.util.List;
import java.util.Map;
public class Community {
    private List<Map<String, String>> persons;
    public void setPersons(List<Map<String, String>> persons)
        { this.persons = persons;
    }
    public List<Map<String, String>> getPersons()
        { return persons;
    }
}
@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController
{ @GetMapping("t11")
    public Community test11() {
        Map<String, String> map1 = new HashMap<String, String>();
        map1.put("firstName", "Ramu");
        map1.put("lastName", "Rao");
        map1.put("address", "BTM");

        Map<String, String> map2 = new HashMap<String, String>();
        map2.put("firstName", "Kumar");
        map2.put("lastName", "Rao");
        map2.put("address", "JP Nagar");

        Map<String, String> map3 = new HashMap<String, String>();
        map3.put("firstName", "Swathi");
        map3.put("lastName", "Kiran");
        map3.put("address", "Jaya Nagar");

        List<Map<String, String>> list = new ArrayList<Map<String, String>>();
        list.add(map1);
        list.add(map2);
        list.add(map3);

        Community community = new Community();
        community.setPersons(list);

        return community;
    }
}
```

React Component:

```
import { useState } from "react"
export default function Test11() {
    const [list, setList] = useState([])
    const callToApi = () => {
        fetch('http://localhost:9090/test/t11')
        .then(response => response.json())
        .then(data => setList(data.persons))
        .catch(error => console.log(error))
    }
    return (<
        {list.map((item, index) =><h1>{Object.entries(item).map([key, value]) =><>{key} : {value}, </>)}</h1>}}
        <button onClick={callToApi}>Get Data</button>
    </>)
}
```

16. Post a simple form in the form of JSON to the serverside API

API:

```
@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController {
```



```

    @PostMapping("t12")
    public Person test12(@RequestBody Person person)
    { System.out.println(person.getFirstName());
      };
      System.out.println(person.getLastName());
      System.out.println(person.getAge());
      return person;
    }
  }
}

React Component:
import { useState } from "react"
export default function Test12(){
  const [firstName, setFirstName] = useState("")
  const [lastName, setLastName] = useState("")
  const [age, setAge] = useState(0);
  const submitHandler = (event) =>
  { event.preventDefault();
    console.log(firstName, lastName, age)
    const person = {firstName, lastName, age};
    console.log(person)
    fetch('http://localhost:9090/test/t12', {
      method: 'POST',
      headers: {
        'Content-Type': 'application/json'
      },
      body: JSON.stringify(person)
    })
  }
  .then(response => response.json())
  .then(data => console.log(data))
  .catch(error => console.log(error))
}

return (<
  First Name : <input type="text"
    value={firstName}
    onChange={(event) => setFirstName(event.target.value)}/><br />
  Last Name : <input type="text"
    value={lastName}
    onChange={(event) => setLastName(event.target.value)}/><br />
  Age : <input type="number"
    value={age}
    onChange={(event) => setAge(event.target.value)}/><br />

  <button onClick={submitHandler} type="button">Submit</button>
</>)
}

```

17. Post a simple form in the form of JSON to the serverside API with validations.

API:

Same as previous

React Component:

```

import { useState } from "react"
export default function Test13(){
  const [firstName, setFirstName] = useState("")
  const [lastName, setLastName] = useState("")
  const [age, setAge] = useState("");
  const [firstNameError, setFirstNameError] = useState("")
  const [lastNameError, setLastNameError] = useState("")
  const [ageError, setAgeError] = useState("")
  const submitHandler = (event) =>
  { event.preventDefault();
    if(firstName === ""){
      setFirstNameError("First Name is required")
    }else{
      setFirstNameError("")
    }
    if(lastName === ""){
      setLastNameError("Last Name is required")
    }
    else{

```

```

        setLastNameError("")
    }
    if(age === ""){
        setAgeError("Age is required")
    }
    else{
        setAgeError("")
    }
    if(firstName !== "" && lastName !== "" && age !==
    0){ console.log(firstName, lastName, age)
    const person = {firstName, lastName, age};
    console.log(person)
    fetch('http://localhost:9090/test/t12', {
        method: 'POST',
        headers: {
            'Content-Type': 'application/json'
        },
        body: JSON.stringify(person)
    })
    .then(response => response.json())
    .then(data => console.log(data))
    .catch(error => console.log(error))
    }
}
return (<
    First Name : <input type="text"
        value={firstName}
        onChange={(event) => setFirstName(event.target.value)}/>
    {firstNameError &&<span style={{color: "red"}}>{firstNameError}</span>}
    <br />

    Last Name : <input type="text"
        value={lastName}
        onChange={(event) => setLastName(event.target.value)}/>
    {lastNameError &&<span style={{color: "red"}}>{lastNameError}</span>}
    <br />

    Age : <input type="text"
        value={age}
        onChange={(event) => setAge(event.target.value)}/>
    {ageError &&<span style={{color: "red"}}>{ageError}</span>}
    <br />

    <button onClick={submitHandler} type="button">Submit</button>
</>)
}

```

18. Post a simple signup form in the form of JSON to the serverside API

API:

```

package com.example.app1;
public class AppUser {
    private String firstName;
    private String email;
    private String password;
    public String getFirstName() {
        return firstName;
    }
    public void setFirstName(String firstName)
    { this.firstName = firstName;
    }
    public String getEmail() {
        return email;
    }
    public void setEmail(String email) {
        this.email = email;
    }
    public String getPassword() {
        return password;
    }
}

```

```

        public void setPassword(String password)
        { this.password = password;
        }
    }

    @RestController
    @RequestMapping("test")
    @CrossOrigin
    public class TestController
    { @PostMapping("t13")
      public AppUser test13(@RequestBody AppUser appUser )
      { System.out.println(appUser.getFirstName());
        System.out.println(appUser.getEmail());
        System.out.println(appUser.getPassword());
        return appUser;
      }
    }
}

React component:
import { useState } from "react";
export default function Test14() {
  const [firstName, setFirstName] = useState("");
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const handleSubmit = (e) => {
    e.preventDefault();
    const person = {firstName, email, password};
    fetch('http://localhost:9090/test/t13',
      {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(person)
      })
    .then(response => response.json())
    .then(data => console.log(data))
    .catch(error => console.error(error));
  }
  return (<
    First Name: <input type="text" value={firstName} onChange={(e) => setFirstName(e.target.value)} /><br />
    Email: <input type="text" value={email} onChange={(e) => setEmail(e.target.value)} /><br />
    Password: <input type="text" value={password} onChange={(e) => setPassword(e.target.value)} /><br />
    <button onClick={handleSubmit} type="button">Submit</button>
  </>)
}

```

19. Post a simple signup form in the form of JSON to the serverside API with a validations.

Please do by yourself

20. Post a simple signup form in the form of FormData to the serverside API

```

API:
@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController
{ @PostMapping("t131")
  public String test13(@RequestParam String firstName,
    @RequestParam String email,
    @RequestParam String password
    ) throws
    Exception{ System.out.println(f
      irstName);
      System.out.println(email);
      System.out.println(password);
      return "success";
    }
}

```

React Component:

```

import { useState } from "react";
export default function Test15() {

  const [firstName, setFirstName] = useState("");
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const handleSubmit = (e) => {
    e.preventDefault();
    const formData = new FormData();
    formData.append('firstName', firstName);
    formData.append('email', email);
    formData.append('password', password);
    fetch('http://localhost:9090/test/t131',
      {
        method: 'POST',
        body: formData
      })
    .then(response => response.json())
    .then(data => console.log(data))
    .catch(error => console.error(error));
  }
  return (<
    First Name: <input type="text" value={firstName} onChange={(e) => setFirstName(e.target.value)} /><br />
    Email: <input type="text" value={email} onChange={(e) => setEmail(e.target.value)} /><br />
    Password: <input type="text" value={password} onChange={(e) => setPassword(e.target.value)} /><br />
    <button onClick={handleSubmit} type="button">Submit</button>
  </>)
}

```

Note:

1. FormData doesn't require Content-Type, it can send any type including binary like a file upload.
2. JSON data should be received through RequestBody in the API
3. FormData should be received through RequestParam in the API

21. Post a simple signup form in the form of FormÚta to the serverside API with file upload.

API:

```

@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController
{ @PostMapping("t132")
  public String test13(@RequestParam String firstName,
    @RequestParam String email,
    @RequestParam String password,
    @RequestPart MultipartFile file) throws Exception{
    System.out.println(firstName);
    System.out.println(email);
    System.out.println(password);
    byte[] bytes = file.getBytes();
    FileOutputStream fout = new FileOutputStream(file.getOriginalFilename());
    BufferedOutputStream out = new BufferedOutputStream(fout);
    out.write(bytes);
    out.flush();
    out.close();
    return "success";
  }
}

```

React Component:

```

import { useState } from "react";
export default function Test16() {
  const [firstName, setFirstName] = useState("");
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [file, setFile] = useState(null);
  const handleSubmit = (e) =>
  { e.preventDefault();
    const formData = new FormData();
    formData.append('firstName', firstName);
    formData.append('email', email);
    formData.append('password', password);

```

```

formData.append('file', file);

fetch('http://localhost:9090/test/t132',
  {
    method: 'POST',
    body: formData
  })
.then(response => response.json())
.then(data => console.log(data))
.catch(error => console.error(error));
}
return (<
  First Name: <input type="text" value={firstName} onChange={(e) => setFirstName(e.target.value)} /><br />
  Email: <input type="text" value={email} onChange={(e) => setEmail(e.target.value)} /><br />
  Password: <input type="text" value={password} onChange={(e) => setPassword(e.target.value)} /><br />
  File: <input type="file" onChange={(e) => setFile(e.target.files[0])} /><br />
  <button onClick={handleSubmit} type="button">Submit</button>
</>)
}

```

Note: If file upload is there then we should use FormData

22. Post a simple signup form in the form with a JSON content and a file

API:

```

@RestController
@RequestMapping("test")
@CrossOrigin
public class TestController
{
    @PostMapping("t15")
    public String test15(@RequestPart Person person,
        @RequestParam MultipartFile file) throws Exception
    {
        byte[] bytes = file.getBytes();
        FileOutputStream fout = new FileOutputStream(file.getOriginalFilename());
        BufferedOutputStream out = new BufferedOutputStream(fout);
        out.write(bytes);
        out.flush();
        out.close();
        System.out.println(person.getFirstName());
        System.out.println(person.getLastName());
        System.out.println(person.getAge());
        return "success";
    }
}

```

React Component:

```

import { useState } from "react";
export default function Test17() {
  const [firstName, setFirstName] = useState("");
  const [lastName, setLastName] = useState("");
  const [age, setAge] = useState("");
  const [file, setFile] = useState(null)
  const handleSubmit = (e) => {
    const person = {firstName, lastName, age}
    e.preventDefault()
    console.log(file)
    const formData = new FormData();
    formData.append("person",
      new Blob([JSON.stringify(person)], { type: "application/json" }));
    formData.append("file", file);
    fetch("http://localhost:9090/test/t15", {
      method: "POST",
      body: formData
    })
  }
  .then(res => console.log(res))
  .catch(err => console.log(err))
}
return (<
  First Name: <input type="text" onChange={e => setFirstName(e.target.value)} /><br />
  Last Name: <input type="text" onChange={e => setLastName(e.target.value)} /><br />
  Age: <input type="text" onChange={e => setAge(e.target.value)} /><br />
  File: <input type="file" onChange={e => setFile(e.target.files[0])} /><br />

```

```
        <button onClick={handleSubmit} type='button'>Submit</button>
      </>
    }
  }
```

23. Post a simple signup form in the form of FormData to the serverside API with validations.
Do by yourself
24. Post a simple Login form in the form of JSON to the serverside API
Do by yourself
25. Post a simple Login form in the form of JSON to the serverside API with validations.
Do by yourself
26. Post a simple Login form in the form of FormData to the serverside API
Do by yourself
27. Post a simple Login form in the form of FormData to the serverside API with a validations.
Do by yourself